LAB 4: Draw the Activity Diagram

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Project: AI Chatbot for University Helpdesk

1. Activity Diagram: Student Login and Chat (RAG) Flow

Shows how a student registers or logs in, asks a question, and receives an answer. The system retrieves relevant chunks from ChromaDB, builds a prompt, and the LLM generates a cited answer. Saving the conversation and displaying the answer run in parallel (fork/join). Questions with no relevant context are logged as unanswered.

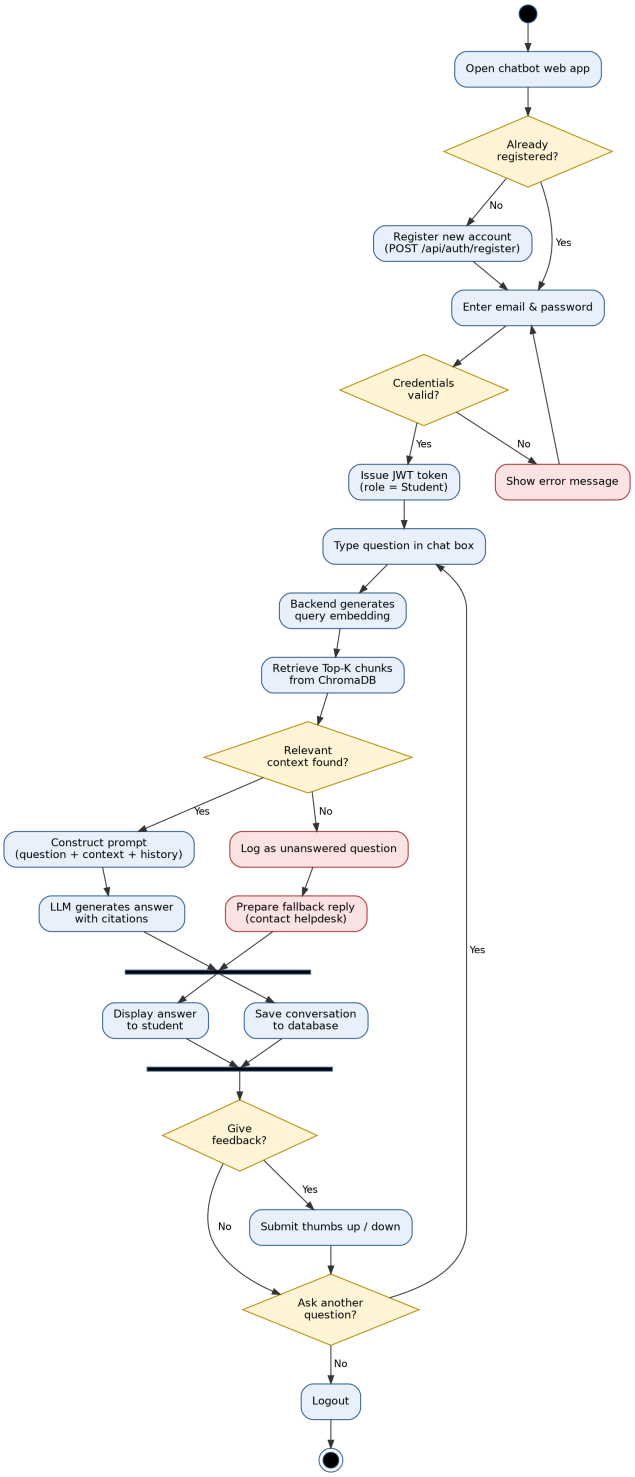


Figure 4.1: Activity diagram for student authentication and chat

2. Activity Diagram: Administrator Knowledge Base Management

Shows the administrator logging in and choosing between uploading a document, deleting a document, viewing analytics, and viewing unanswered questions and feedback. Uploaded files are validated, split into chunks, embedded, and stored in ChromaDB and SQLite.

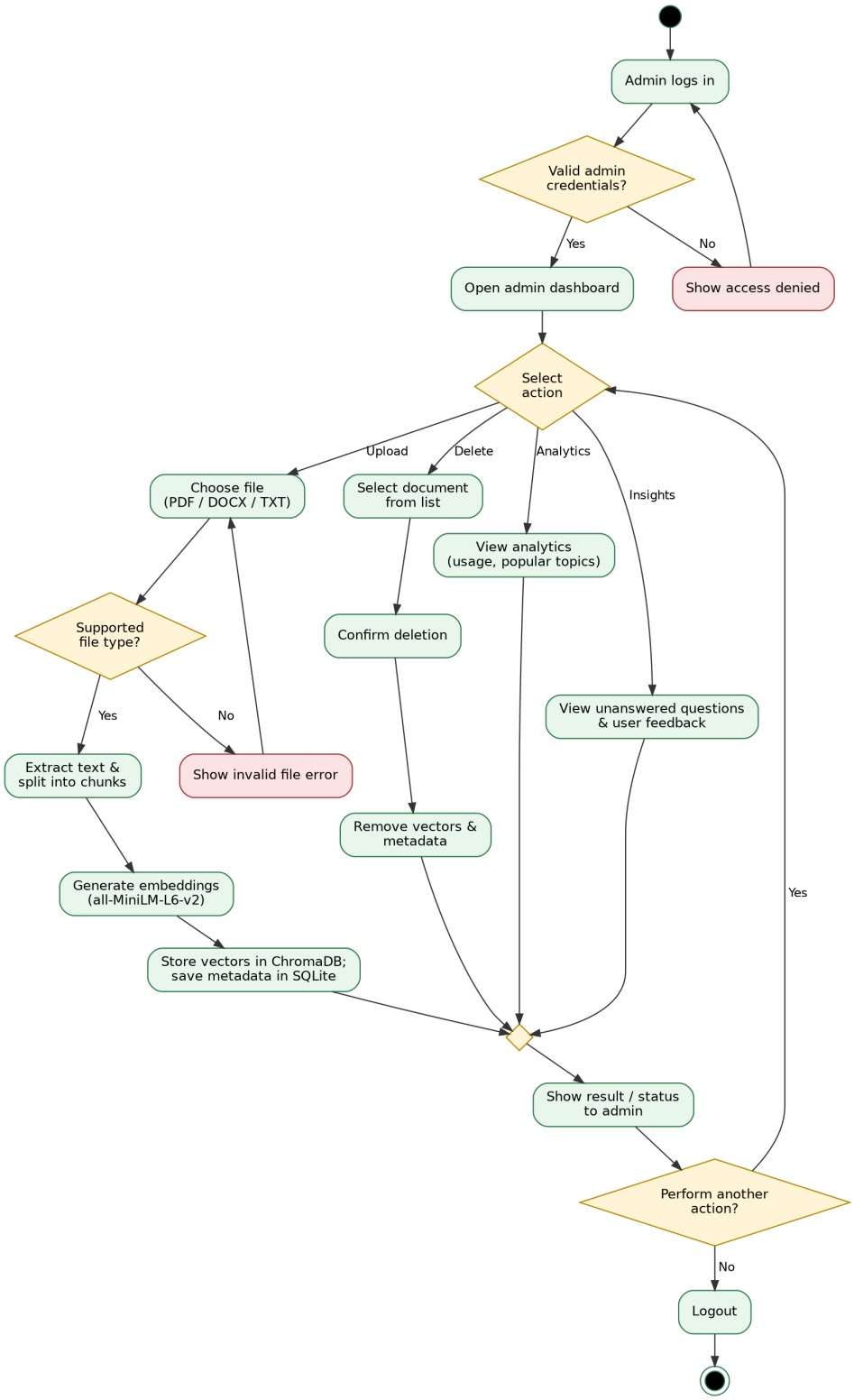


Figure 4.2: Activity diagram for administrator operations